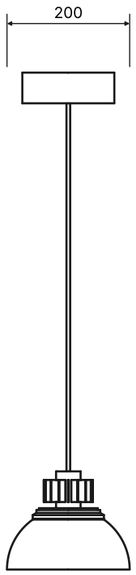
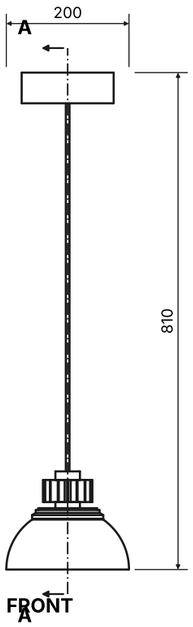
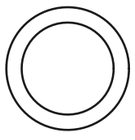
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

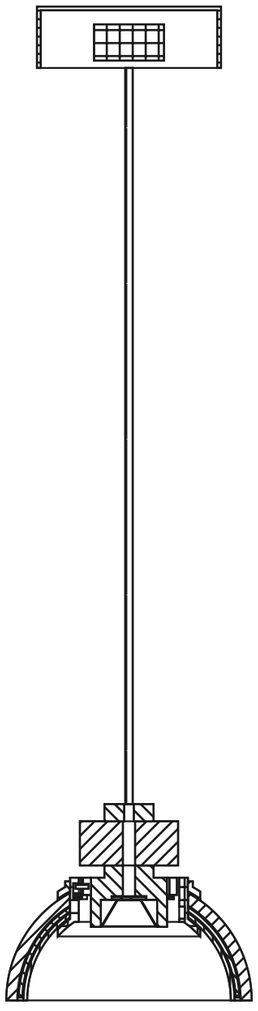
General arrangement

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



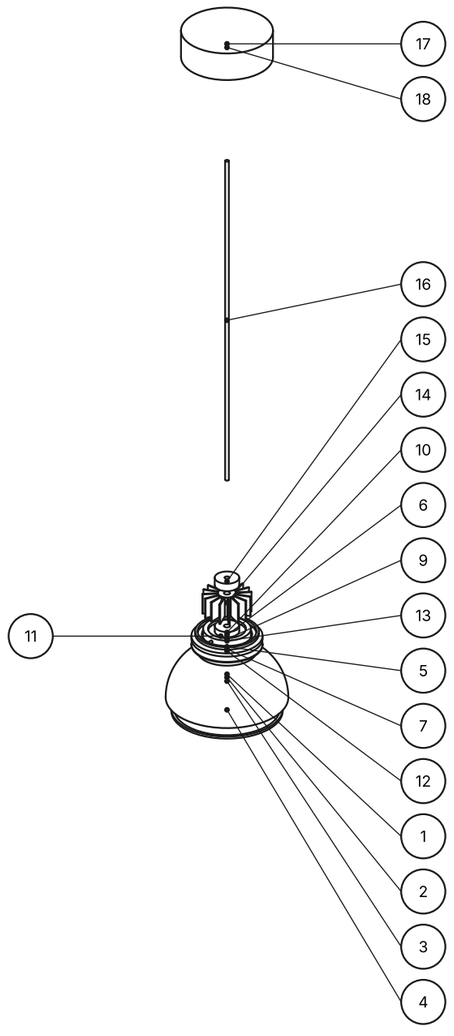
Section A-A

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

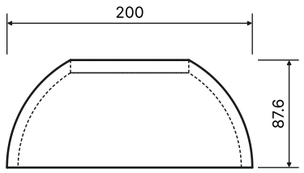
mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

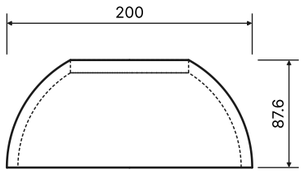
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: formed, acrylic-formed; 446 cm3 each; 1 for the set; 1:5

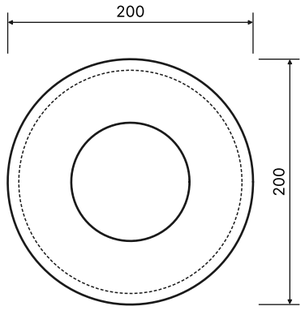
the outside: 10 mm tinted cast sheet drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds (9 at the crown, thinner toward the rim: the former confirms); its outer face never touches the mould, so it keeps the cast sheet's gloss



FRONT



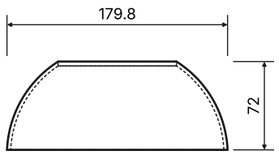
FROM THE RIGHT



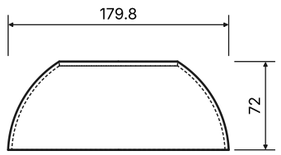
FROM ABOVE

2. liner: formed, acrylic-opal-formed; 117 cm3 each; 1 for the set; 1:5

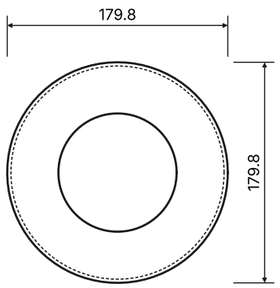
the diffuser: 3 mm volume-diffusing (opal) cast acrylic, light-diffusing grade, about 70 % transmission drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds. Opal scatters through its volume, so it diffuses wherever it touches the tinted layers (a bead-blasted clear liner goes clear where it is bonded or wetted)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

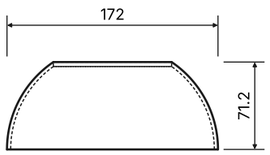
mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

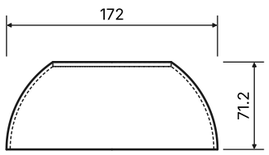
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. skin: formed, acrylic-formed; 111 cm3 each; 1 for the set; 1:5

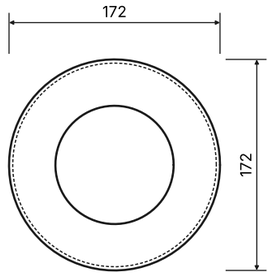
the inside: 3 mm tinted cast sheet formed into a female mould, so its inner (seen) face keeps its gloss and the opal is seen only through tint



FRONT



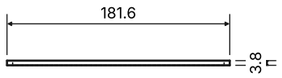
FROM THE RIGHT



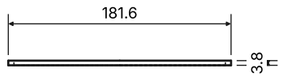
FROM ABOVE

4. rim: lathe, cast-acrylic; 10 cm3 each; 1 for the set; 1:5

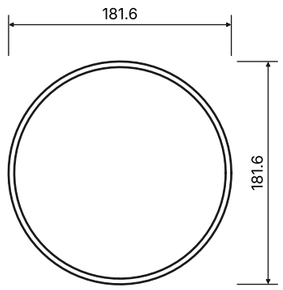
closes the three layers at the mouth, bonded and polished with them; a press-in ePTFE vent in its face keeps the gaps between the layers at room pressure and dry



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

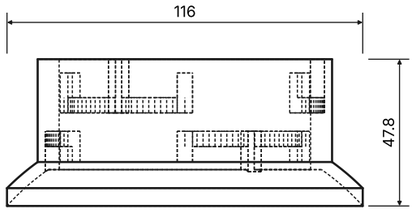
mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

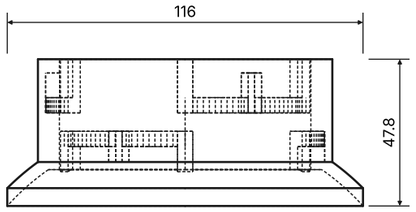
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. collar: lathe, aluminium; 84 cm3 each; 1 for the set; 1:2

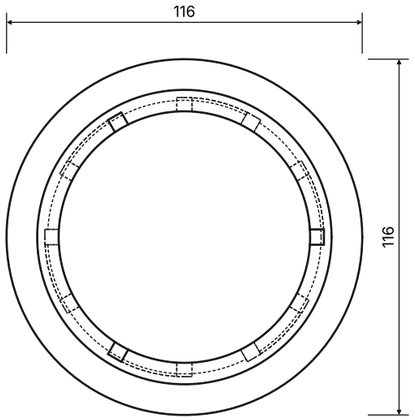
aluminium 6061, black anodised, satin: a tube through the crown with a spherical flange under the skin; three blind L-slots with T-ended notches (the quarter-turn bayonet, 30 deg) work either way up; M96 × 1.5 outside for the clamp ring. It carries the cover mechanically: the bond is a seal only



FRONT



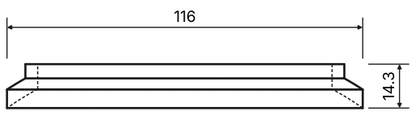
FROM THE RIGHT



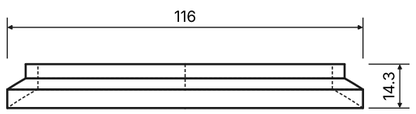
FROM ABOVE

6. clamp: lathe, aluminium; 24 cm3 each; 1 for the set; 1:2

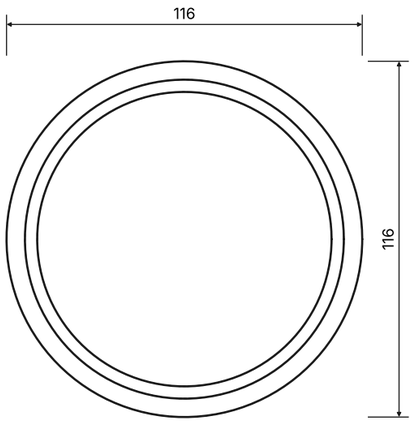
aluminium, black anodised: screwed onto the collar over the crown, its spherical face on a silicone ring, to a torque the shade maker sets (the acrylic is clamped between two silicone rings, never metal on acrylic); a drop of thread-lock



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

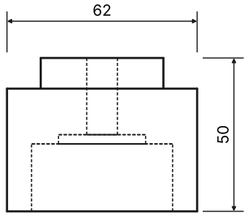
mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

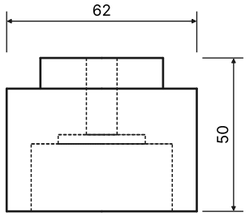
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

9. hub: lathe, aluminium; 93 cm3 each; 1 for the set; 1:2

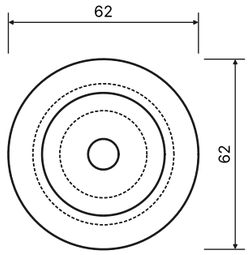
aluminium 6061, black anodised: the core, turned from bar. The COB on its holder at the top of the 46 bore (22 up: the cut-off, about 46 deg from the vertical), the reflector in the bore, the PEEK ring round it, the fins bolted on its neck on a thermal pad; a 10 bore for the wires



FRONT



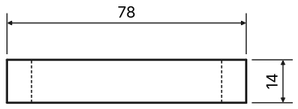
FROM THE RIGHT



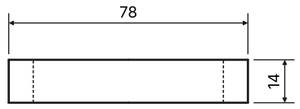
FROM ABOVE

10. ring: lathe, peek; 25 cm3 each; 1 for the set; 1:2

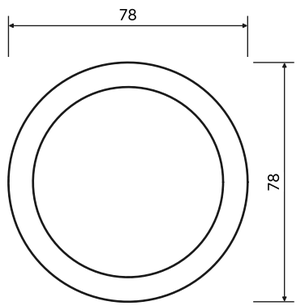
PEEK, unfilled (0.25 W/m K), black: the thermal break. It carries the three pins and the detent, so the cover touches the fixture only through it; 4 to 5 mm of air round the core inside the collar



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

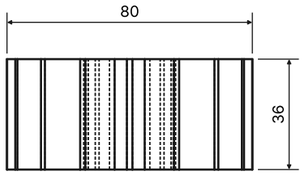
mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

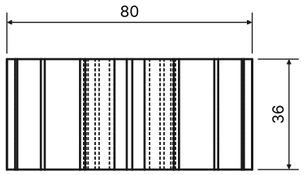
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

14. fins: extrusion, aluminium; 49 cm3 each; 1 for the set; 1:2

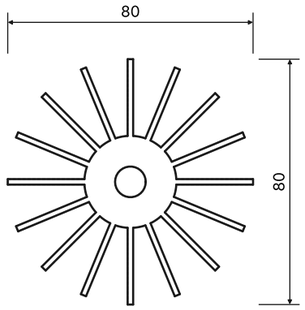
the heat sink, in room air above the cover: 36 cut from a standard radial (sunflower) extrusion, 80 across, 16 fins, the cut faces turned, black anodised (the black also radiates: emissivity about 0.85); concentric with the cap and the hub



FRONT



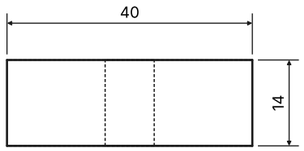
FROM THE RIGHT



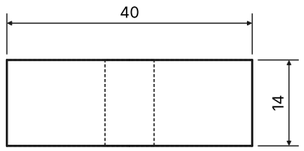
FROM ABOVE

15. gland: lathe, cast-acrylic; 17 cm3 each; 1 for the set; 1:1

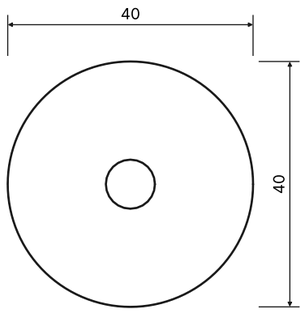
the cap: black cast acrylic, polished, over the listed cord grip (the strain relief, tested with the fixture's weight), on two M3 into the fins' core



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

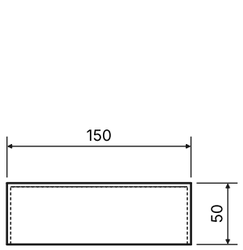
mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

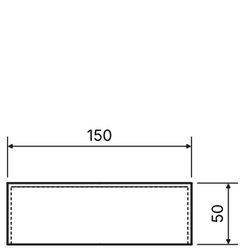
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

17. canopy: lathe, cast-acrylic; 118 cm3 each; 1 for the set; 1:5

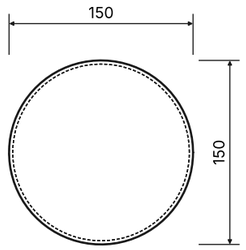
150 × 50, black cast acrylic, polished: over the driver and a steel junction-box strap (the strap carries the load)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

mm | issue 2026-10-25 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	446 cm3, 210 × 210 × 98	1,534.41
2	liner	formed	1	Cast acrylic sheet, volume-diffusing (opal), formed over an aluminium mould	117 cm3, 190 × 190 × 82	1,212.73
3	skin	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	111 cm3, 182 × 182 × 81	1,212.30
4	rim	lathe	1	Cast acrylic tube, tinted through	192 × 192 × 14	65.26
5	collar	lathe	1	Aluminium 6061	126 × 126 × 58	67.96
6	clamp	lathe	1	Aluminium 6061	126 × 126 × 24	54.65
7	gaskets	buy	1	Two moulded silicone rings (Shore 40, black, 1 mm, spherical): one between the collar's flange and the skin inside, one between the clamp ring and the shell outside		3.00
8	vent	buy	1	Gore or alike A press-in ePTFE pressure-equalising vent (about 4 across), in the rim ring: it lets the air gaps between the layers breathe and keeps out dust and liquid		1.50
9	hub	lathe	1	Aluminium 6061	72 × 72 × 60	52.78
10	ring	lathe	1	PEEK rod, unfilled	88 × 88 × 24	110.05
11	pins	buy	1	Three stainless dowel pins 4 × 8 (ISO 8734), pressed into the hub: the bayonet's pins		1.50
12	reflector	buy	1	Ledil or a spinner (to choose) A faceted reflector in high-reflectance aluminium (Alanod Miro kind, about 95 %) for a Vero 10, 45 across at its mouth, 21 deep, inside the hub's bore: it turns the light that the black bore absorbed (about half		4.00
13	cob	buy	1	Bridgelux Vero SE 10 COB, 2700 K, CRI 90 (BXRC-27G1000-D-73), on an Ideal Chip-Lok holder with a thermal pad, at the top of the hub's bore. DS120: at 350 mA 26.0 V typical, 9.1 W, about 1,130 lm at Tc 25 C and 1,020 at Tc 85 C; at 700 mA (its		12.00
14	fins	extrusion	1	Aluminium 6061	90 × 90 × 46	14.32
15	gland	lathe	1	Cast acrylic tube, tinted through	50 × 50 × 24	47.40
16	cord	buy	1	any (UL listed) Black textile-braided pendant cord, 2-conductor 18 AWG, UL listed (SVT-rated, or a listed Class 2 cord: past the driver in the canopy it carries the LED's 26 V DC, not mains), 3 m cut to the drop; a listed cord grip in the cap and		16.00
17	canopy	lathe	1	Cast acrylic tube, tinted through	160 × 160 × 60	106.44
18	driver	buy	1	to choose A Class 2 constant-current driver at about 175 mA, 22 to 30 V window, 120 V AC in, phase-cut dimmable, FCC Part 15 Class B (consumer), dimming to 10 % or below, PF 0.90 or more at its load (JA8): the Vero SE 10 at about 175 mA runs		22.00
19	plunger	buy	1	Kipp or Misumi A stainless ball spring plunger M5, light force (about 10 N), threaded into the hub: the bayonet's detent, clicking into a dimple in the collar		4.50
20	listing	buy	1	UL or Intertek (ETL) Listing the pendant to UL 1598 (and the table lamp's battery and charger to UL 153 / IEC 62368): a one-off cost a design, spread over the first run		[PRICE]

Parts list

Hemisphere: Pendant in cast acrylic, its shades swapping by hand

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. Hemisphere (the Dome until round 35; Hemisphere North in rounds 35 to 37): the shade hangs mouth down. Its shades swap by hand: lift the shade 5 mm, turn it 30 degrees and lower it off. To fit one, line up the three slots, push up, turn until the plunger clicks, and let go: its weight holds each pin up in its notch, so a knock cannot turn it off. No tools, and the shade is never live (the COB, the hub and the driver stay up).
2. The shade is clamped, not hung on a glue joint: the collar's flange sits under the crown inside, a ring screws down over it outside, the acrylic between two silicone rings. A failed bond cannot drop it.
3. The wall is tint, opal, tint: a 9 mm tinted shell, a 3 mm volume-diffusing opal liner, a 3 mm tinted skin, each formed from cast sheet on an aluminium mould, nested, closed at the mouth by a tinted rim with a vent. Opal scatters through its depth, so it still diffuses where it touches the tint (a bead-blasted clear liner would go clear wherever it was bonded). Nothing white is seen.
4. Light: a Bridgelux Vero SE 10 (2700 K, CRI 90) at the top of a 46 bore lined with a reflector, 22 up: the cut-off about 46 deg from the vertical, and about 85 % of its light out (the black bore passed 52 %). Two levels, the owner to choose: about 1,100 lm at the COB on a Fulham T1T11200350-15L (350 mA, FCC Part 15B consumer; the COB runs 26 V, 9.1 W; the Mean Well PCD-16-350A the bar-only alternate, FCC Part 18 non-consumer), or about 575 lm at about 175 mA (4.3 W, driver to source). Dimmed by a phase-cut (TRIAC) dimmer; a 0-10 V driver for bars that want it.
5. Heat: the hub's fins sit above the shade in room air (an 80 radial extrusion, black); a PEEK ring carries the bayonet's pins, so the hub reaches the shade only through it and 4 to 10 mm of air. Estimated at 1,100 lm in a 35 C room: the hub 52 C, the collar and the acrylic beside it 42 C (hemisphere_thermal.py; to measure, README round 35).
6. Mains stops at the canopy: the driver is there, and the cord carries the LED's 26 V DC (Class 2). The cord is a listed 18 AWG 2-conductor in a black textile braid, its grips tested with the fixture's weight. Listed to UL 1598 with any shade.
7. Hang over a bar at 750 to 900 centres, the rim 700 to 800 above the counter; over a table, one a table, the rim about 650 to 750 above it.

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